

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY MODULE 3
UNIT: NETWORK DESIGN AND MANAGEMENT

UNIT CODE: 0612 451 07A

TIME: 4 HOURS

TOTAL 50 MARKS

Instructions to Candidates

- ✓ *Read the whole paper before you begin.*
- ✓ *Attempt **all** tasks. Tasks A to F are practical; Task G is a short oral/written check.*
- ✓ *You may use a simulator (e.g. Cisco Packet Tracer) or physical equipment, as directed by your examiner.*
- ✓ *Diagrams must be neat, ruled and clearly labelled. Hand-drawn or software-drawn diagrams are both accepted.*
- ✓ *Call the examiner to witness each working stage **[DEMO]** before moving on.*
- ✓ *Mobile phones and notes are not allowed..*

Scenario

A small company, **BrightPath Consultants**, has moved into a new office with three departments: **Human Resources (HR)**, **Accounts**, and **Reception**. Each department has **one computer**. The company has **one USB printer** that must be shared by all departments, and an **internet connection** from an ISP (supplied through a modem/router).

You have been hired as the network technician. You must plan, install, configure and test a network that meets the requirements below.

Equipment Provided

Item	Quantity
Desktop computers (HR-PC, ACC-PC, REC-PC)	3
USB printer (with USB cable)	1
Switch (8 or 24 port)	1
Router / wireless router with internet access (ISP simulated)	1
UTP Cat5e/Cat6 straight-through patch cables	5
RJ45 connectors, crimping tool, cable tester (cabling task), cable stripper	1 set
Diagram sheets	1

Network Requirements

Functional Requirements

The installed network **must**:

ID	Requirement
FR1	Interconnect the three departments (HR, Accounts and Reception) so that all computers can communicate with each other.
FR2	Allow all three departments to use ONE shared USB printer. The printer is connected by USB to one computer (the print host, REC-PC) and shared over the network so that HR and Accounts can also print.
FR3	Allow all three departments to share internet access through a single connection.
FR4	Allow file sharing between departments, with a shared folder that all three computers can access.

FR5	Give every device a unique IP address in the same private address range, with a default gateway for internet access.
FR6	Allow Accounts to keep a folder that ONLY Accounts and HR can access (Reception must be denied access).
FR7	Allow users to verify connectivity using standard tools (ping, ipconfig/ifconfig, tracert).

Non-Functional Requirements

The network **should**:

ID	Quality	Requirement
NFR1	Reliability	Use a wired star topology so the failure of one cable affects only one device.
NFR2	Performance	Use at least 100 Mbps links (Fast Ethernet or better) on all connections.
NFR3	Security	Protect router/switch management with passwords, change default credentials, and (if wireless is used) secure it with WPA2 or better.
NFR4	Scalability	Use an addressing plan that leaves room for at least 20 more devices, and a switch with spare ports.
NFR5	Manageability / Usability	Use meaningful device names, label cables and ports, and keep documentation up to date.
NFR6	Maintainability	Use a consistent, documented addressing scheme (static addresses for the router and the computers).
NFR7	Cost-effectiveness	Use only the equipment provided; no unnecessary additional devices.

Tasks

Task A: Physical Network Diagram

Draw a **well-labelled physical diagram** of the network. Your diagram must show:

- All three computers (HR-PC, ACC-PC, REC-PC), the USB printer, the switch, the router and the internet/ISP cloud.
- The USB cable connecting the printer to its host computer (REC-PC).
- Cable type used on each link (e.g. Cat6 UTP straight-through).
- Switch port numbers and router interface names used for each connection.
- A title, a legend/key, your name, and the date.

Task B: Logical Topology Design

1. Draw the **logical topology** showing how data flows between devices: IP addresses, the subnet, the default gateway, the path to the internet, and how HR-PC and ACC-PC reach the printer through the print host (REC-PC).
2. State the **physical topology** and the **logical topology** used (e.g. star physical, broadcast logical) and give **two reasons** for your choice.

Task C: Addressing Table

Using the private network **192.168.10.0/24**, prepare an addressing table with these columns:

Device	Interface / Port	IP Address	Subnet Mask	Default Gateway	Static / DHCP

Rules:

- Include the router (LAN and WAN), switch (management, if applicable), HR-PC, ACC-PC and REC-PC. (The USB printer has no network address; show it on the physical diagram only.)

- The router LAN interface must be the **first usable address**.
- The router must use a **static** address.
- Reserve a block for future growth and state it clearly.
- State the network address, broadcast address, number of usable hosts and CIDR notation.

Task D: Installation and Configuration

1. **Cabling:** Connect all devices according to your diagram. Label each cable and port. If crimping, make and test one straight-through cable and show the tester result.
2. **Router configuration:** Set the LAN IP, enable DHCP (if used), enable NAT/internet sharing, set the router hostname, and set a management password.
3. **Computer configuration:** Rename the computers (HR-PC, ACC-PC, REC-PC), assign IP settings per your table, and set the same workgroup (e.g. BRIGTHPATH).
4. **Printer sharing:** Connect the USB printer to REC-PC, install its driver, and share it. Add the shared printer on HR-PC and ACC-PC, then print a test page from each department.
5. **File sharing:** Create a shared folder named “CompanyShare” accessible to all three computers. Create a second folder, “FinanceHR”, accessible only to Accounts and HR users. Prove that Reception is denied access.
6. **Internet sharing:** Show that all three computers can browse the internet (or reach the simulated ISP server) through the shared connection.

Task E: Testing and Verification

Complete the test log below and demonstrate each test to the examiner.

No.	Test	Action / Command	Expected Result	Actual Result	Pass / Fail
1	HR-PC pings ACC-PC	ping	Replies received		
2	ACC-PC pings REC-PC	ping	Replies received		
3	REC-PC pings HR-PC	ping	Replies received		
4	Each PC pings default gateway	ping	Replies received		

5	Shared printer visible on HR-PC and ACC-PC	Add printer / browse \\REC-PC	Printer listed		
6	Each PC reaches internet/ISP	ping / browse	Success		
7	Trace route to external address	tracert	First hop = gateway		
8	Print a test page from each PC	Print	Page prints		
9	Access CompanyShare from each PC	Open \\<IP>	Accessible		
10	Reception opens FinanceHR	Open \\<IP>	Access denied		

Also record the output of **ipconfig /all** (or **ifconfig**) for **one** computer.

Task F: Troubleshooting

The examiner will introduce **one fault** into your working network (for example: wrong subnet mask, disconnected cable, wrong gateway, or the print host being switched off). You must:

1. Identify the symptom and describe the troubleshooting steps you followed.
2. State the cause of the fault.
3. Fix the fault and re-test.

Task G: Short Questions

1. What is the purpose of the default gateway?
2. Why must the computer connected to the USB printer stay switched on for other departments to print?
3. Name two ways to improve the security of this network.